Crop Diversity and Drought Resilience

We tested whether countries that grow a wider mix of crops lose less agricultural production in droughts, using 31 years of data for 107 countries.

Pranav Swarup Kumar · Prathmesh Sharma · Pragya Mishra · Gursahib Singh · Arushi Tandon

The question and what we add

Does growing a wider mix of crops make a country’s agricultural production more resilient to drought?

Four faces of resilience

Stability, resistance, recovery and vulnerability, each measured separately.

Response diversity

How differently a country’s crops react to drought, not only how many it grows.

Cropland-weighted drought

Drought averaged over where crops grow, not over whole countries.

107

countries

157

crops

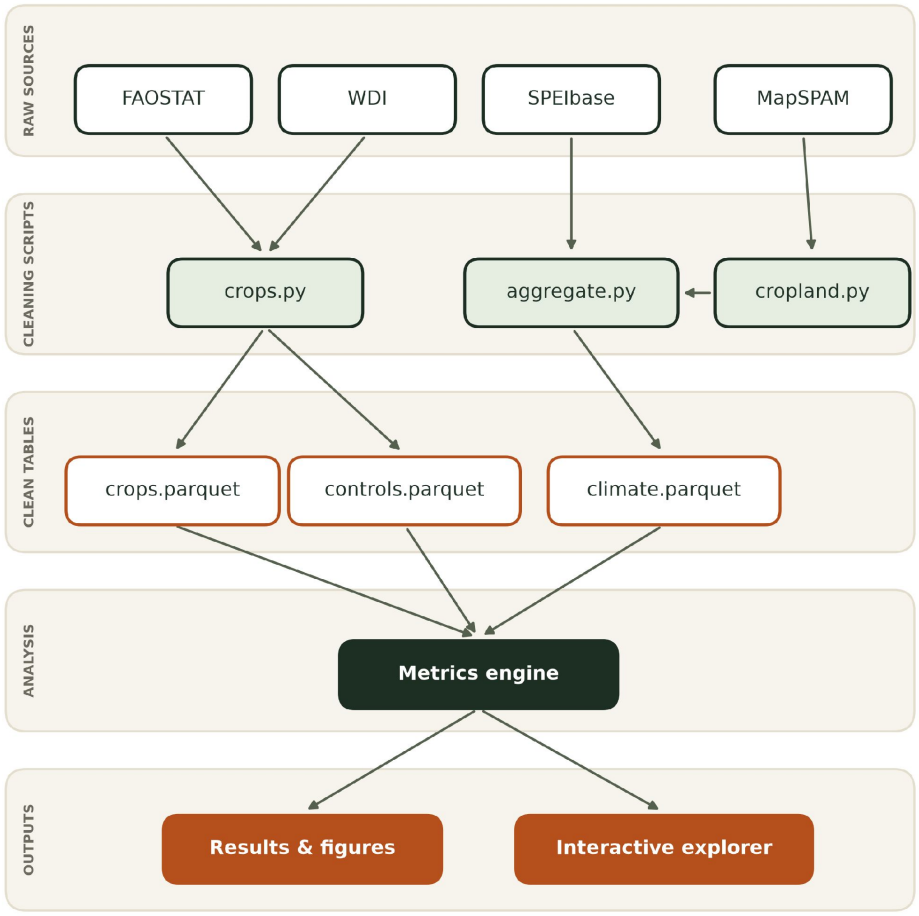
1993–2023

31 years

232

drought events analysed

From four public datasets to country results



Source	Used for
FAOSTAT	Crop area, production and value
SPEIbase v2.10	Drought index on a 0.5° grid
MapSPAM 2020	Where each of 46 crops grows
World Bank WDI	Income and fertiliser use

Every step is a script that checks its own output. The results feed the figures in this deck and an interactive country explorer.

Also used: Natural Earth admin-0 boundaries for country shapes. FAOSTAT supplies QCL (production), QV (value of production) and RL (land use).

What each dataset contributes

FAOSTAT

Crop production and value, 1993–2023

QCL gives area harvested and production, QV the gross production value in constant 2014–16 international dollars, RL land use. After removing 11 aggregate items: 267,838 country-crop-years, 195 countries, 157 crops. 41% of rows carry an estimated or imputed flag. Where no value is published (3.8% of rows, 1.9% of value), we use production × the item's price.

SPEIbase v2.10

Drought index, monthly on a 0.5° grid to December 2023

SPEI is precipitation minus potential evapotranspiration, standardised, so 0 is normal and −1 or below is drought. We use the 12-month index at December and the 6-month index, averaged per country and year: 7,130 country-years for 230 countries.

MapSPAM 2020 v2

Where 46 crops grow, 5 arc-minute grid

Physical harvested area, all technologies, summed to 1,274 million ha on the 0.5° grid (38,162 cells with cropland). It is one snapshot, used as the weight for the whole window, so we assume cropland does not move much between 1993 and 2023.

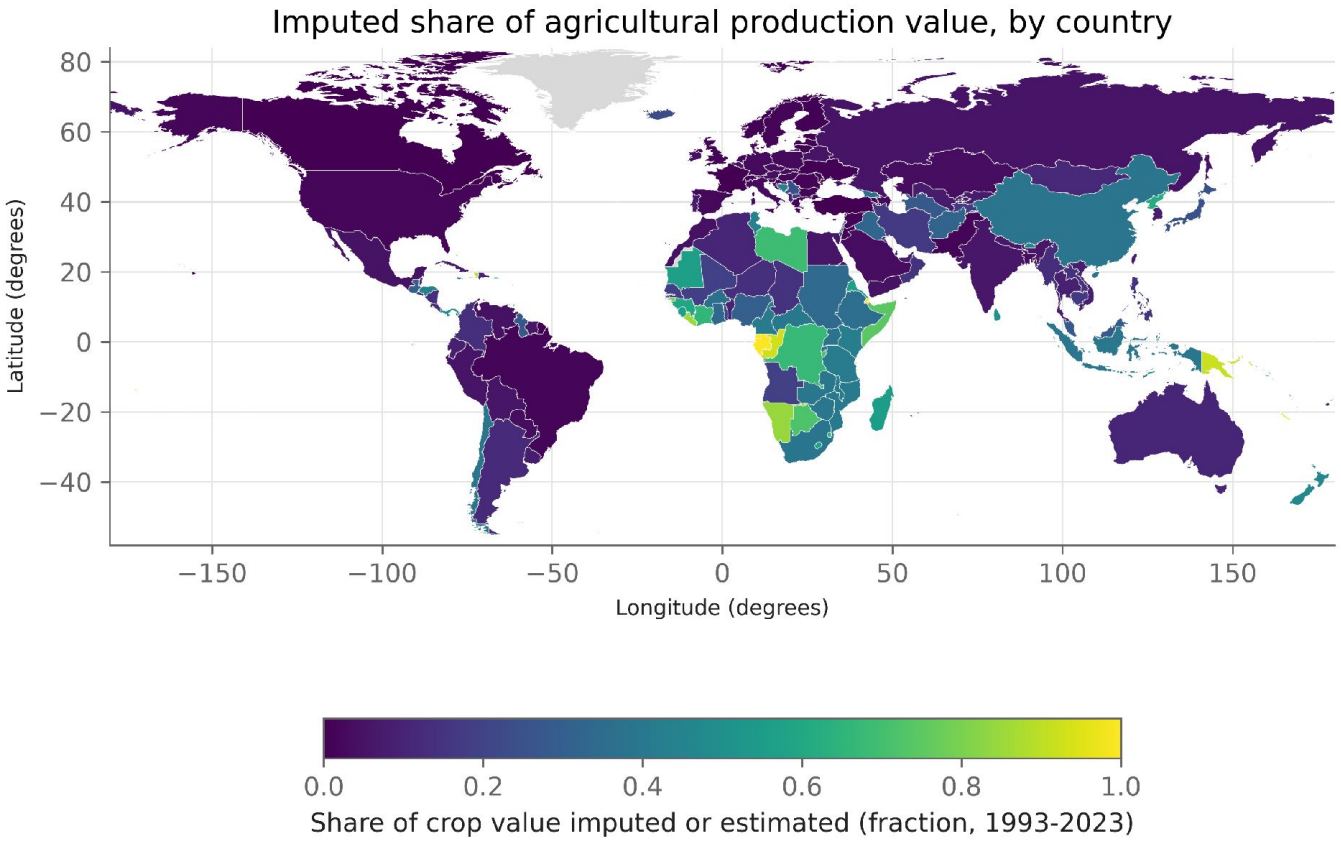
World Bank WDI + FAOSTAT RL

Control variables for the regressions

GDP per capita and fertiliser use (kg/ha) from WDI; irrigation share is irrigated land divided by cropland from FAOSTAT. Coverage is 95%, 91% and 90% of country-years. Merged entities (BLX, SCG, SDX) and Taiwan have no WDI series.

Country shapes and merged entities. Natural Earth 1:10m boundaries are dissolved to our codes. Belgium–Luxembourg (BLX), Serbia–Montenegro (SCG) and Sudan (SDX) are merged to keep series continuous across 1993–2023.

Which countries have data we can trust



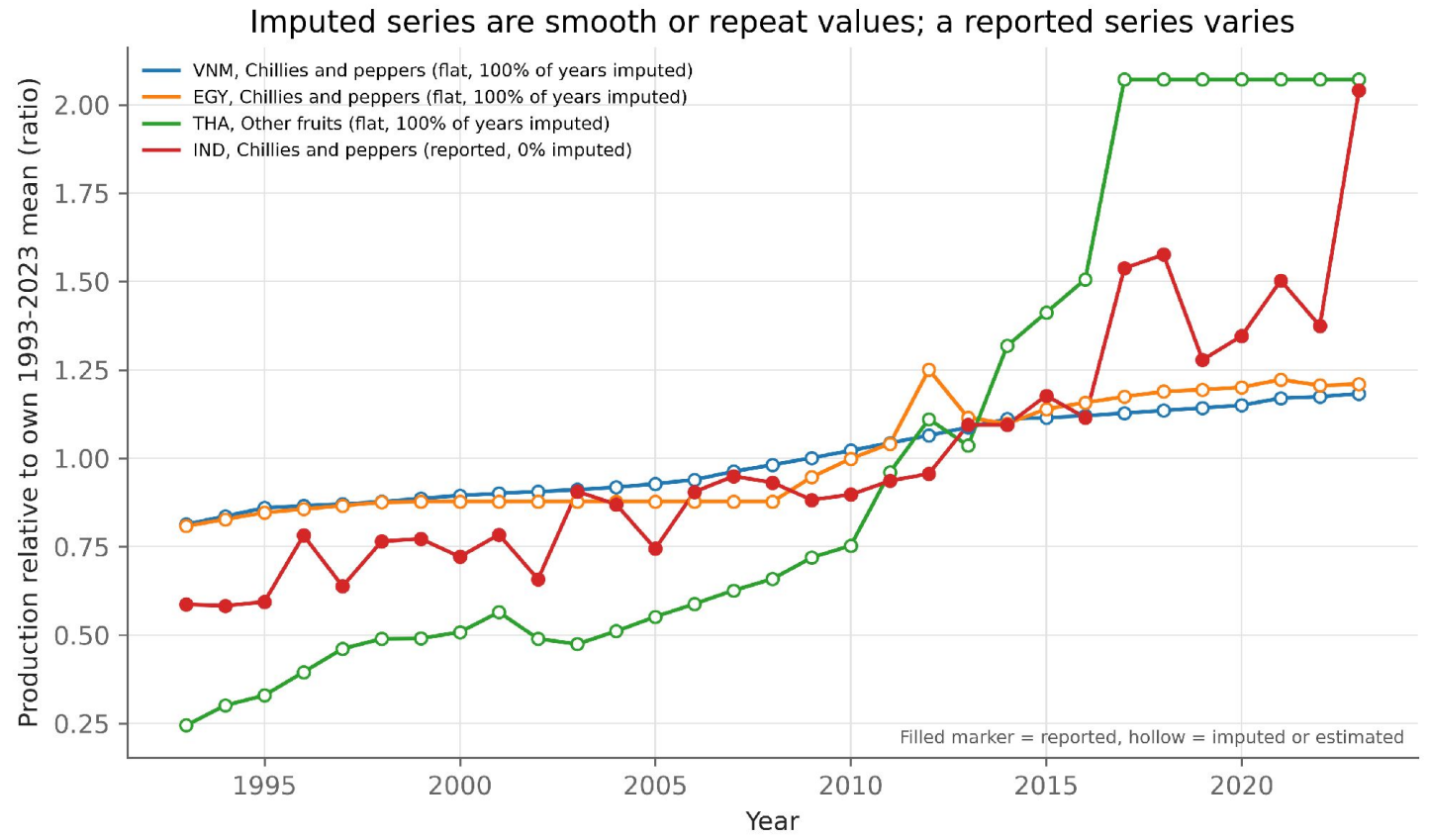
Source: FAOSTAT QCL and QV (bulk download), flags E and I = estimated or imputed. Grey = no data.

107

of 195 countries included

A country needs 100,000 ha harvested, 5 crops and 25 years of mostly reported data. The map shades each country by the share of its crop value that FAOSTAT marks as estimated or imputed (dark = mostly reported); grey countries failed the rules. The 107 included countries hold 91% of global crop value.

Why imputed data had to be screened



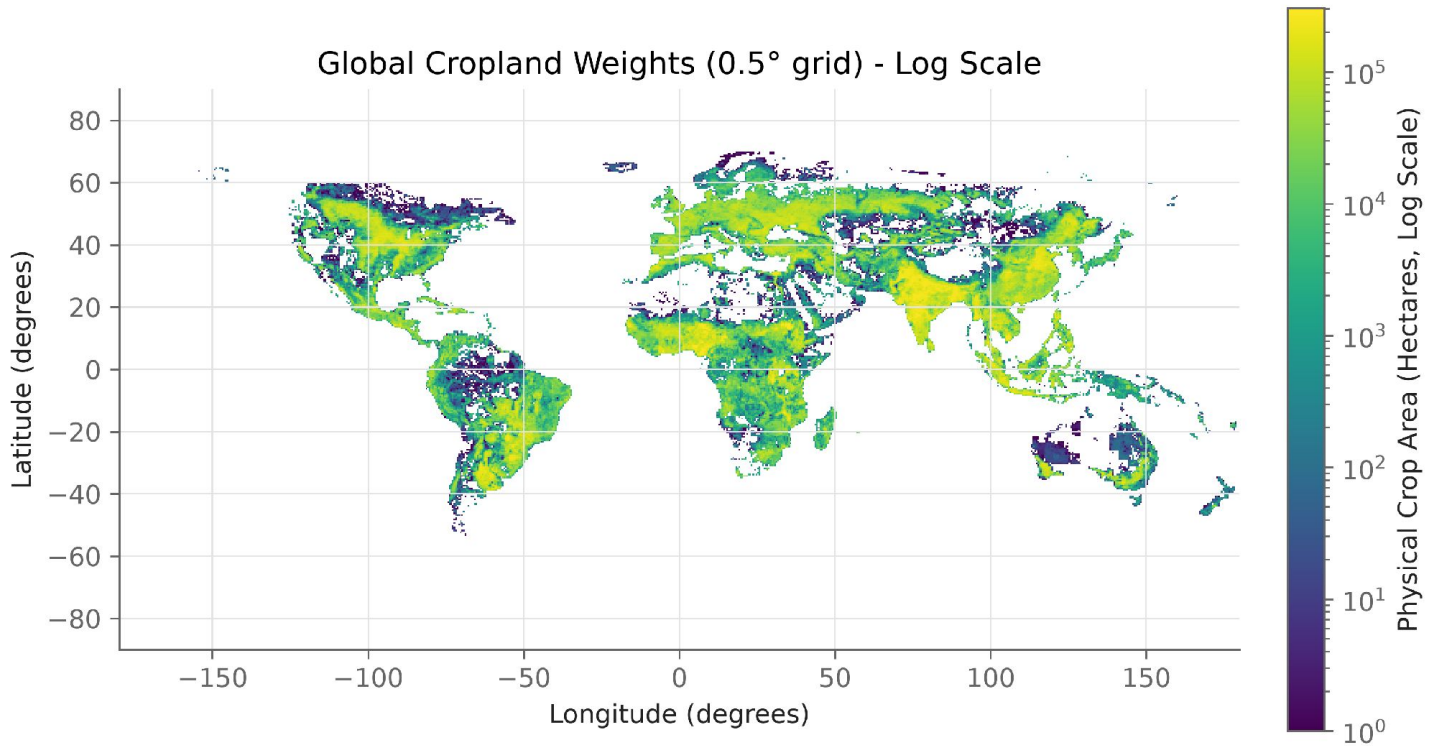
Source: FAOSTAT QCL and QV (bulk download), flags E and I = estimated or imputed.

719

crop series flagged as flat

Each line is one crop’s production relative to its own mean; hollow markers are imputed years. The imputed series run smooth or repeat one value, which would look like stability when nothing was measured. We keep them in national totals but fit crop drought responses on reported years only.

Where the world’s crops grow



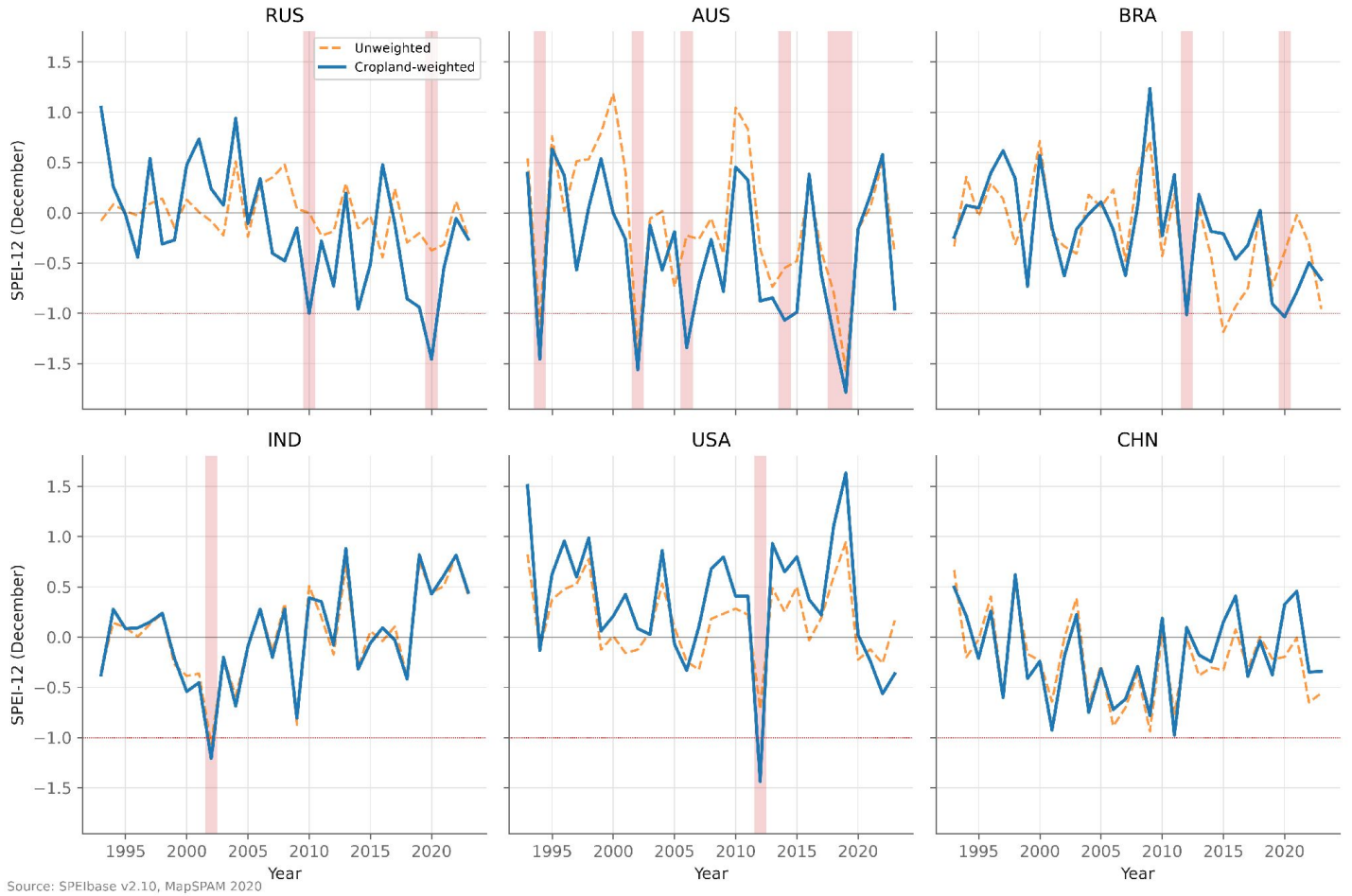
Source: MapSPAM 2020 v2

1,274

million hectares of cropland

Each 0.5° grid cell is coloured by the physical area of 46 MapSPAM crops (log scale); 38,162 of 259,200 cells hold cropland. This area is the weight in every drought average, so deserts and ice count for nothing.

Drought measured over cropland, not whole countries

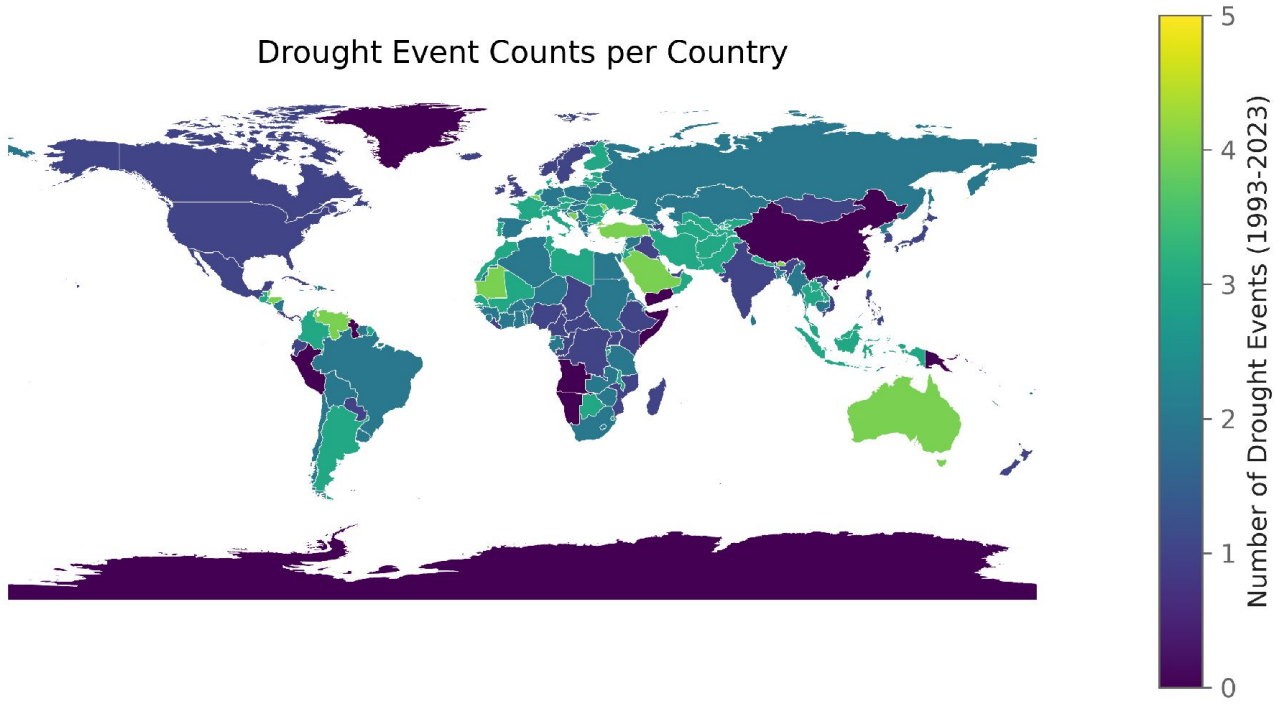


SPEI-12

drought index, cropland-weighted

Blue is the drought index averaged where crops grow; orange is the same index averaged over the whole country; shaded years have SPEI-12 at or below −1. The two differ most where farmland is concentrated, as in Russia and Australia. In the USA only the cropland-weighted line marks 2012 as a drought.

Drought events, 1993–2023



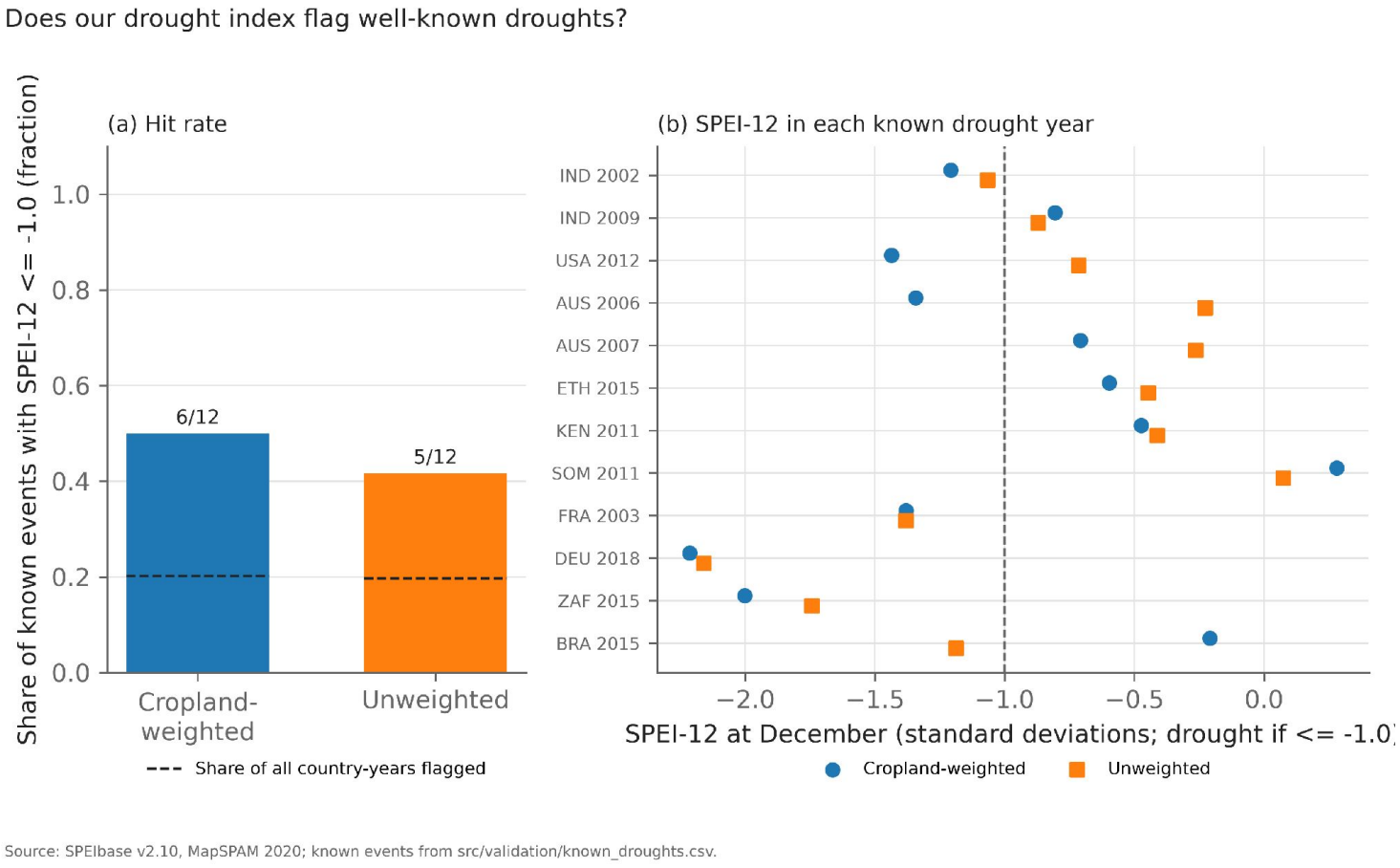
Source: SPEIbase v2.10, MapSPAM 2020, Natural Earth

431

events in 187 countries

An event is the first year of a run of drought years (December SPEI-12 ≤ −1), kept only if the three years before were drought-free and three years after fall inside the window. The map counts events per country; 232 events fall in the 107 countries analysed.

Check against 12 documented droughts



Source: SPEIbase v2.10, MapSPAM 2020; known events from src/validation/known_droughts.csv.

6 / 12

flagged with cropland weights

Left: share of 12 documented droughts flagged; the dashed line is the share of all country-years flagged (about 20%), so both versions beat chance. Right: SPEI-12 in each drought year, flagged when left of −1. Whole-country SPEI flags 5 of 12; allowing ±1 year the weighted index reaches 7 of 12. East Africa 2011 and Brazil 2015 are missed.

How we calculate: diversity and the value index

1 Crop diversity, hill1

$$hill1 = exp(-\Sigma\, p_i \ln p_i)$$

p_i = crop i's share of mean harvested area, 1993–2023. It is the effective number of crops: it equals the crop count if all share area equally and falls when a few crops dominate. India: 18.9; median 9.2.

2 National value, y□

$$y\square = \Sigma\, value_const \text{ over crops}$$

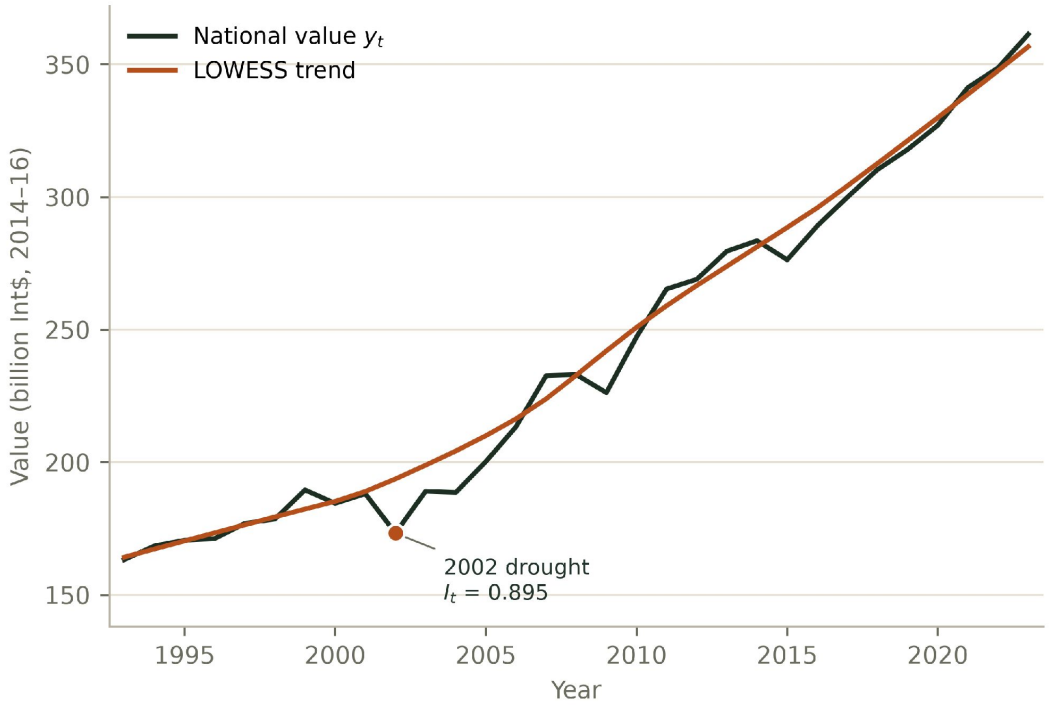
Gross production value in constant 2014–16 international dollars, so crops can be added. Tonnes are never summed across crops.

3 Detrend to get I□

$$I\square = y\square / trend\square$$

Trend is LOWESS (frac 0.5) on 1993–2023. I□ = 1 is on trend, 0.9 is 10% below. Anomaly a□ = (y□ − trend□) / trend□.

India: national crop value and its LOWESS trend



The dip below the trend in 2002 gives I□ = 0.895. Everything that follows is measured on I□, so long-run growth from better inputs is not mistaken for resilience.

Source: FAOSTAT QV; India series from outputs/metrics_national.parquet

The four faces of resilience: how each is calculated

Stability

$1 / CV$

CV = sd(y□ − trend□) / mean(y□)

Higher means smaller swings around the trend. Uses all 31 years.

India 34.7
median 13.0

Resistance

$I□ / mean(I□_{-3} \dots I□_{-1})$

Median over drought events; t = event year

1 means no loss in the drought year; below 1 is a loss against the three years before.

India 0.89
median 0.98

Recovery

$mean(I□_{+1} \dots I□_{+3}) / I□$

Median over drought events

Above 1 means the next three years rebound beyond the drought-year level.

India 1.05
median 1.02

Vulnerability

share of years with $I□ < 0.9$

Loss of more than 10% below trend

Higher means more bad years. vuln_drought counts only years with SPEI-12 ≤ −1.

India 0.03
median 0.10

Which years count. Resistance and recovery use only drought events: 232 events in 102 countries, each with three drought-free years before and three years after inside 1993–2023. Countries with no usable event get no value. Stability and vulnerability use every year, so they describe chronic instability rather than one shock.

Worked example: India’s 2002 drought

India national value index I□, 1996–2006



Event year t = 2002

SPEI-12 = -1.21, with no drought in 1999–2001, so 2002 starts an event. I□ = 0.895.

Resistance

Before: (1.040 + 0.996 + 0.995) / 3 = 1.010

0.895 / 1.010 = 0.89

Recovery

After: (0.951 + 0.924 + 0.954) / 3 = 0.943

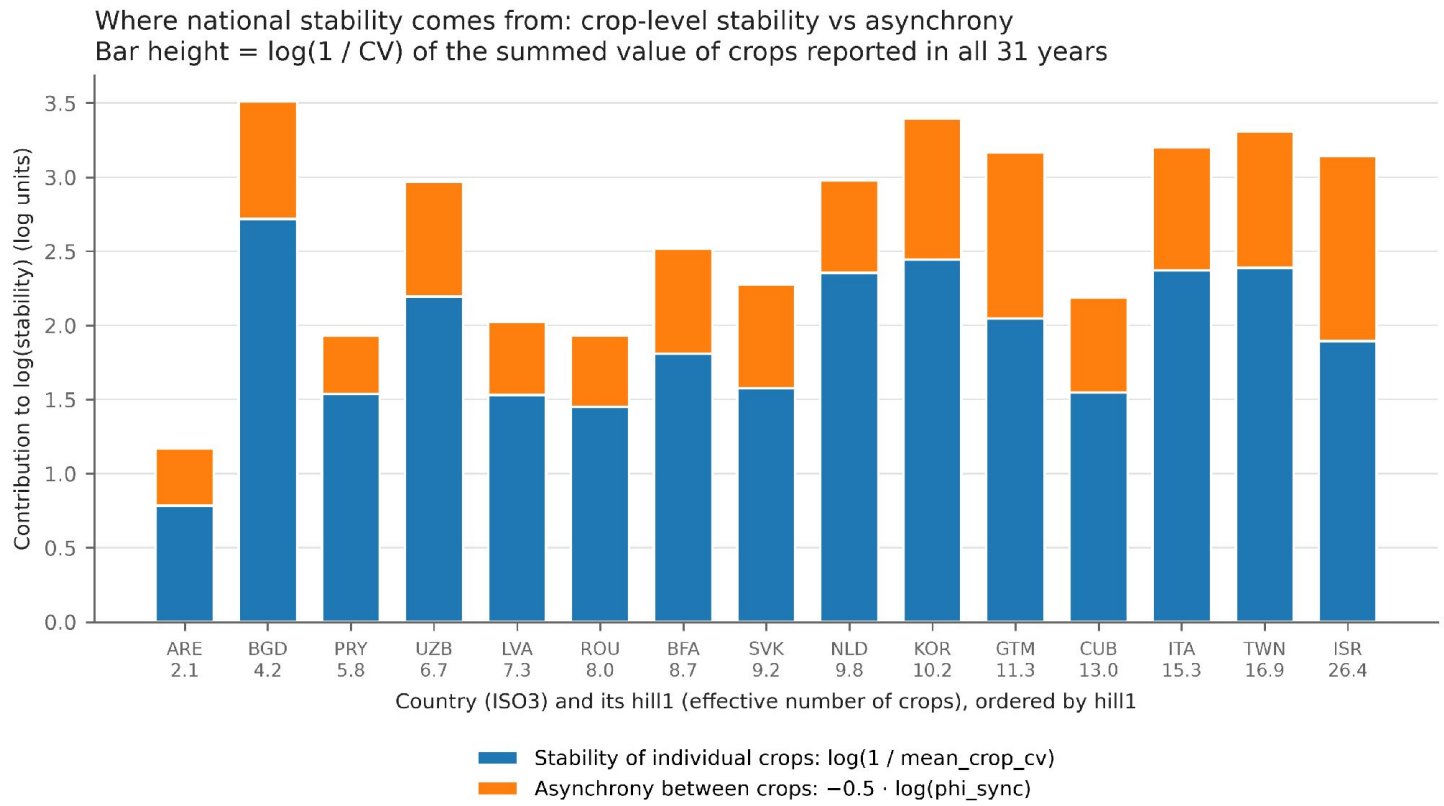
0.943 / 0.895 = 1.05

Vulnerability

2002 is the only year of 31 below 0.9, so 1 / 31 = 0.03. It was a drought year, so vuln_drought = 1 / 1 = 1.0.

India has one usable event, so its resistance and recovery rest on this single drought.

Where national stability comes from



Source: FAOSTAT (QCL, QV), SPEIbase v2.10, MapSPAM 2020, World Bank WDI. 1993–2023, n = 107 countries.

0.23

median synchrony between crops

Each bar is a country's log stability, split into two parts. Blue: how steady its individual crops are. Orange: how out of step they are (on a 0–1 scale where 1 means all crops move together). Both add to stability, and in the more diverse countries, such as Israel, the out-of-step part is a larger share.

How we test the link between diversity and resilience

Rank correlation

Spearman ρ between each predictor (hill1, resp_div) and each face, with 95% bootstrap intervals from 2,000 resamples of countries. n = 107 for hill1.

Regression with controls

OLS with outcome and predictors standardised, so a coefficient is the change in SD per SD. Controls: log GDP per capita, fertiliser use and irrigation share (window means). Robust HC3 errors; n = 96–101, as merged entities lack WDI series.

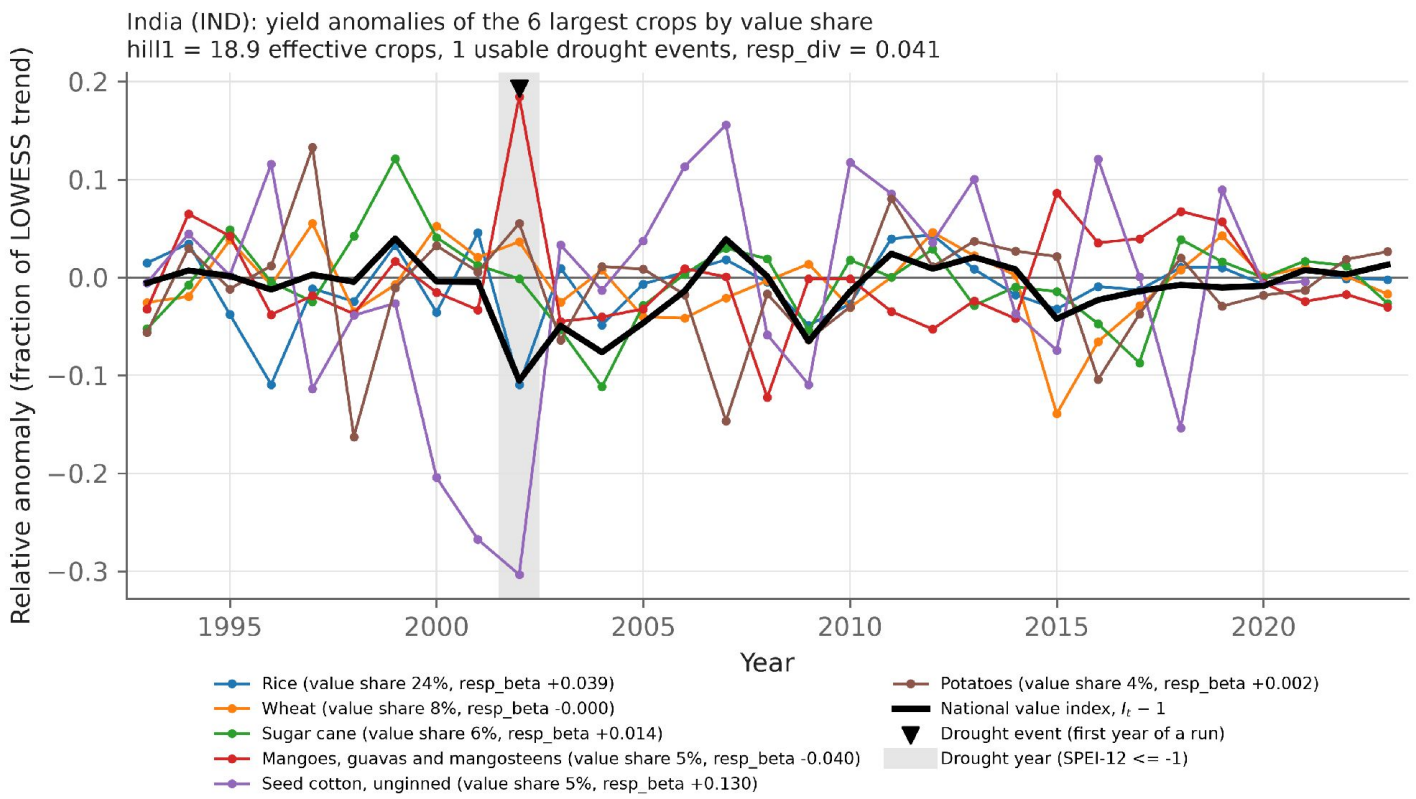
Robustness

Repeat the correlations with LOWESS frac 0.3 and 0.7, severe droughts only (≤ −1.5), countries with at most 10% imputed value, and each world region left out in turn.

Extra checks

Response-diversity paradox: resp_div against the most sensitive single crop. Spatial dilution: country area against SPEI variance. Flash droughts: 6-month against 12-month index versus yield anomalies.

One country in detail: India



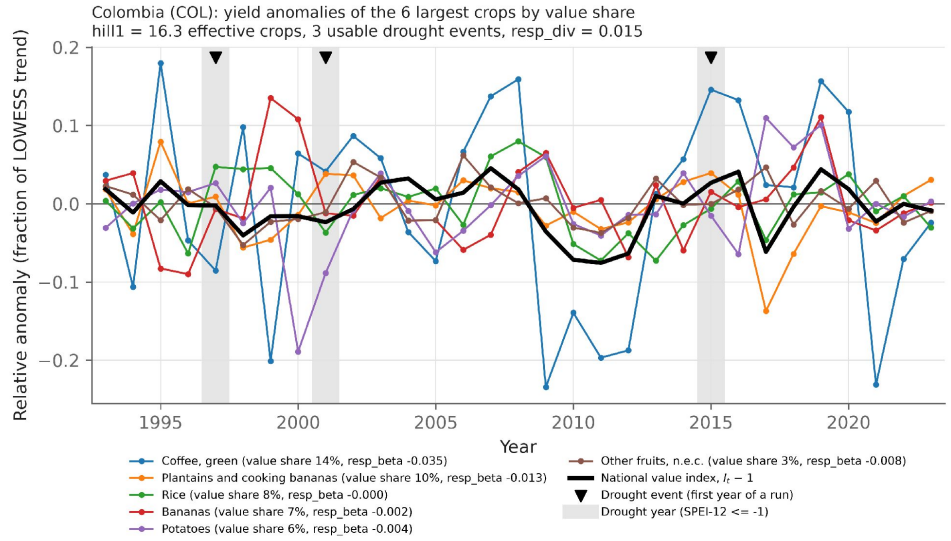
Source: FAOSTAT (QCL, QV), SPEIbase v2.10, MapSPAM 2020. 1993–2023. Crop lines use non-imputed years only.

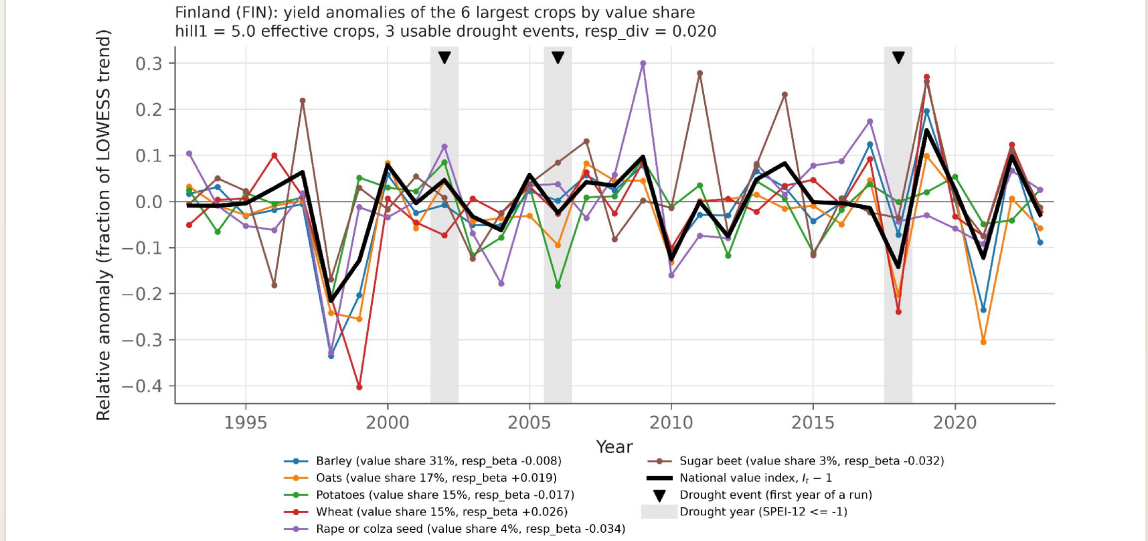
−10%

national value against trend, 2002

Thin lines are the yield anomalies of India's six largest crops; the thick black line is the national value index minus 1; the grey band is the drought year. In 2002 cotton yield fell about 30% and rice about 11%, while mango yield rose, so the national index ended about 10% below trend. India grows 18.9 effective crops, and the spread of responses cushioned the loss.

Two contrasting countries: Colombia and Finland





Colombia: 16.3 effective crops

Stability 29.3 · synchrony 0.06 · bad years 0 of 31

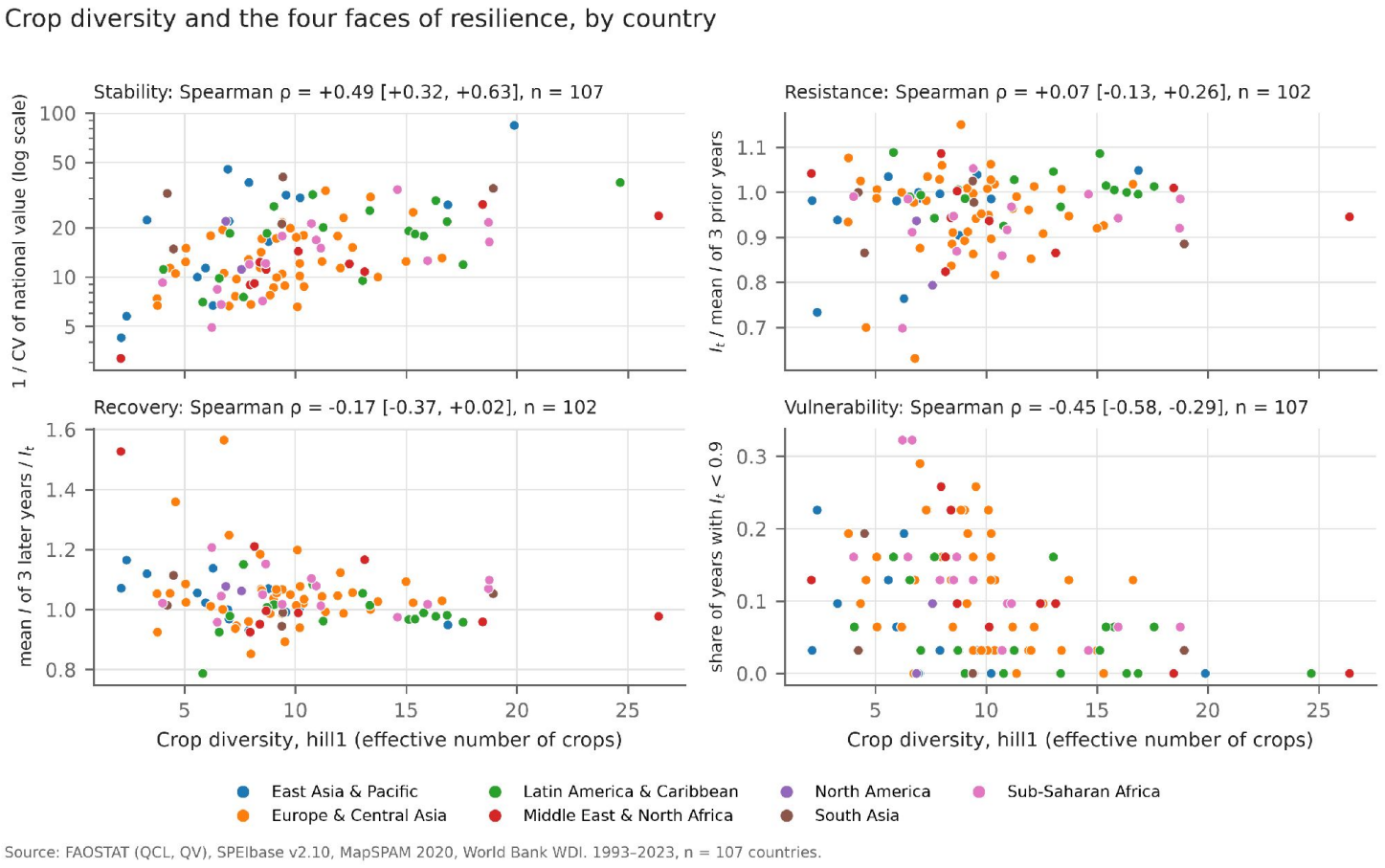
Coffee swings between about −23% and +18% of trend, yet the national value index (thick black line) never falls more than about 8% below trend, because other crops move out of step. The three drought events (triangles) leave no clear mark on the national line.

Finland: 5.0 effective crops

Stability 12.3 · synchrony 0.36 · bad years 5 of 31

Barley, oats, wheat and potatoes tend to fall in the same years, so the national line follows them. The deepest dip, 1998 at about −22%, is not a drought year; in the 2018 drought the national index fell about 14% below trend, with wheat down about 24%.

Crop diversity and the four faces of resilience

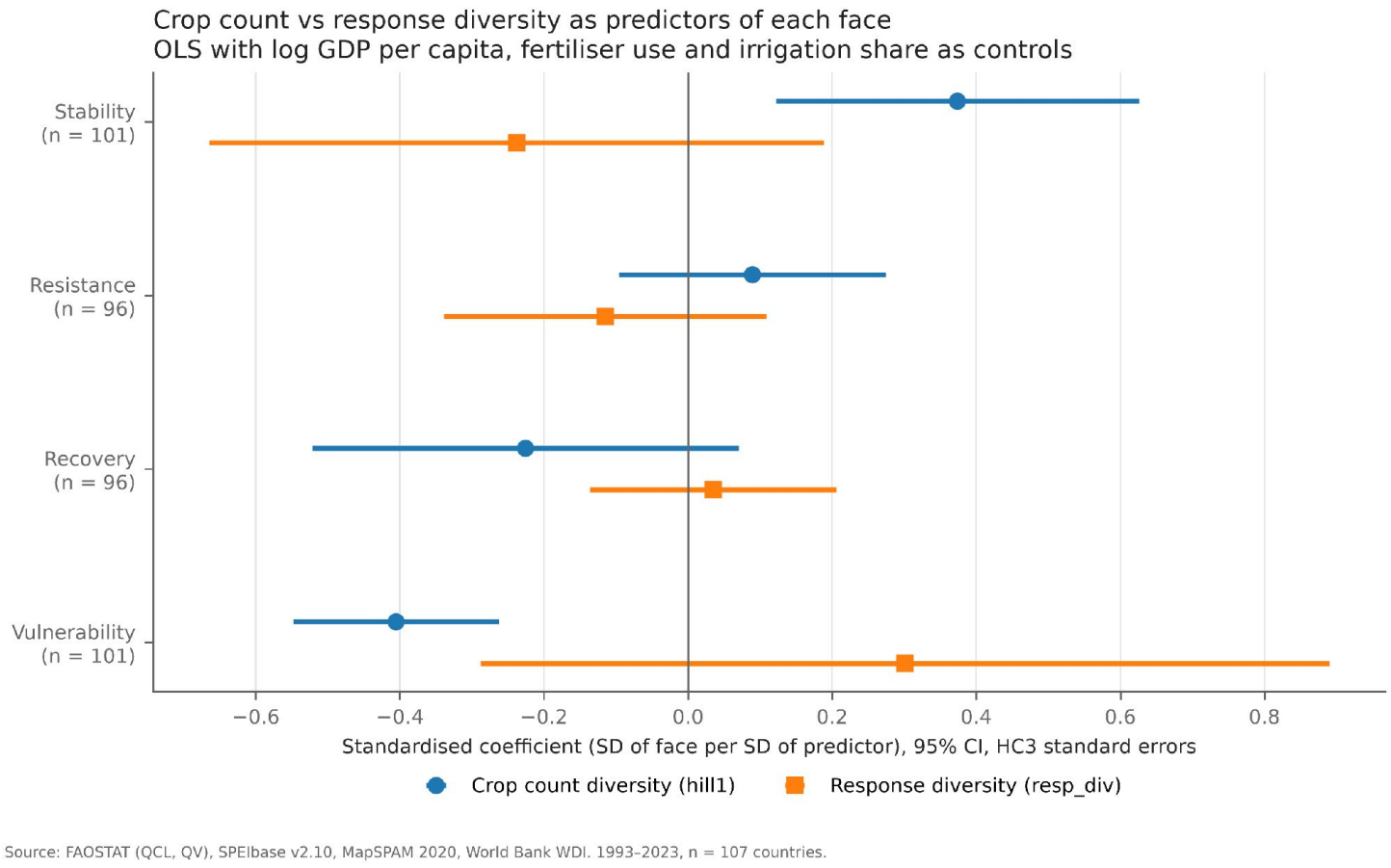


+0.49

diversity and stability

Each dot is a country, coloured by region; the x-axis is crop diversity (hill1). Stability rises with diversity (+0.49) and the share of bad years falls (−0.45). Resistance (+0.07) and recovery (−0.17) show no clear link: their intervals include zero and the dots form a flat cloud.

Crop count against response diversity



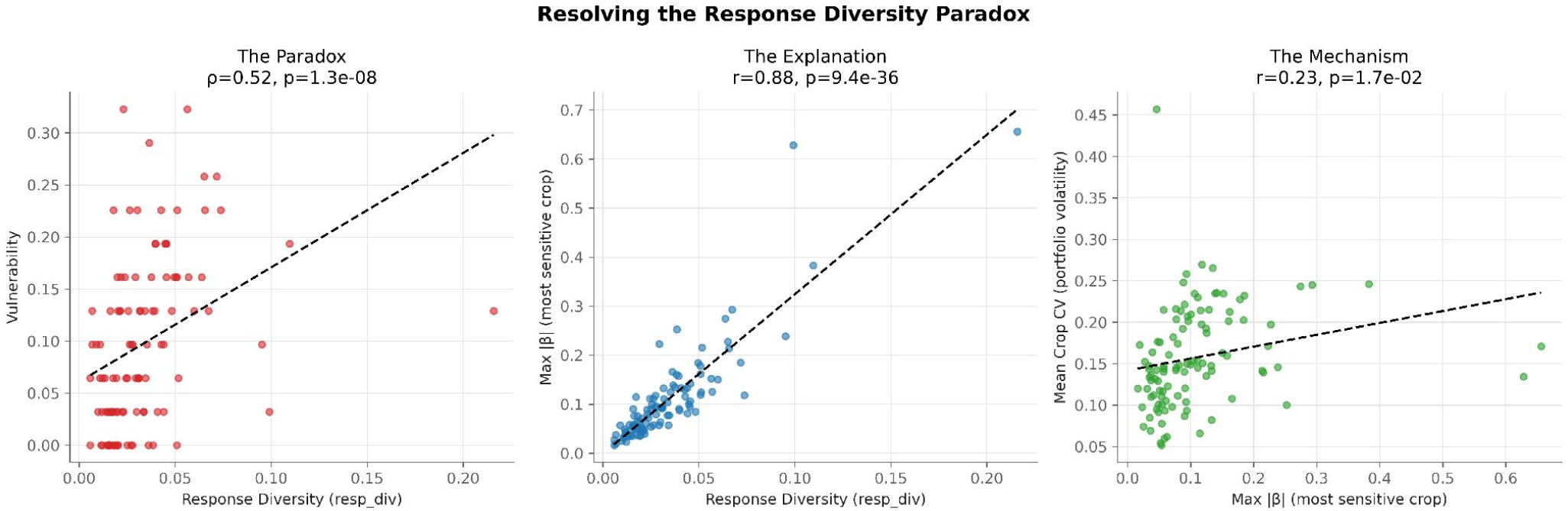
Source: FAOSTAT (QCL, QV), SPEIbase v2.10, MapSPAM 2020, World Bank WDI. 1993–2023, n = 107 countries.

−0.41

vulnerability per SD of crop diversity

Dots are standardised regression coefficients with 95% intervals; an interval crossing zero means no clear effect. With income, fertiliser and irrigation held constant, crop diversity still predicts stability (+0.37) and fewer bad years. Response diversity does not predict stability or overall vulnerability, but it goes with more losses in drought years (+0.36).

Why response diversity points the other way



+0.52

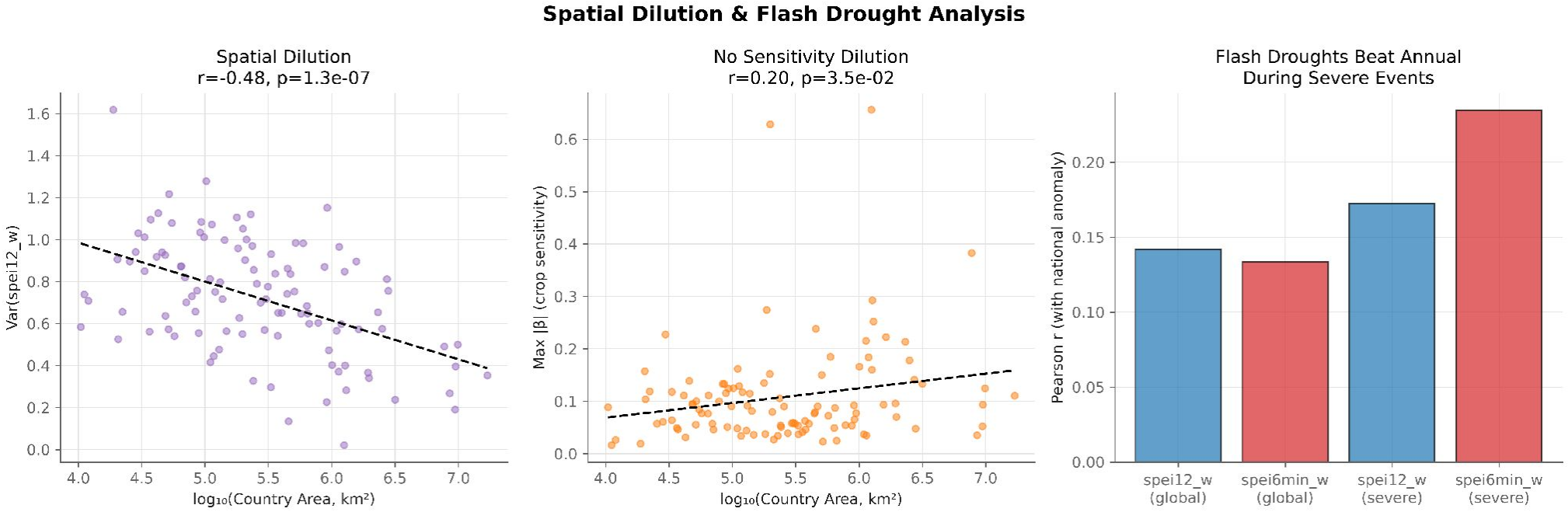
response diversity and vulnerability

0.88

response diversity and the most sensitive crop

Left: countries with higher response diversity have more bad years, the opposite of the insurance idea. Middle: the score tracks the single most drought-sensitive crop almost one for one (r = 0.88). Right: that crop also makes the portfolio more variable. A high score mostly marks one hyper-sensitive crop, not a balanced mix of responses.

Country size and short droughts



−0.48

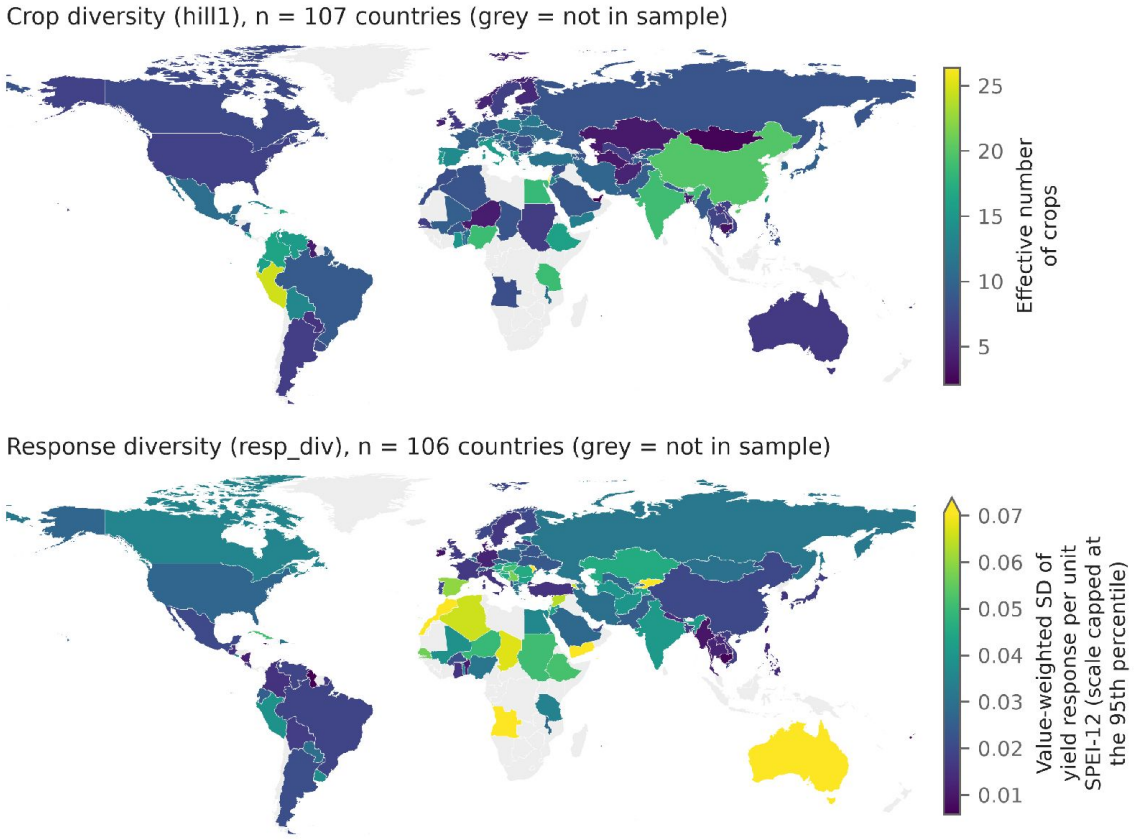
country area and spread of the drought index

0.23 vs 0.17

six-month vs twelve-month index, severe years

Left: the bigger the country, the flatter its averaged drought index, because dry and wet regions cancel. Middle: crop sensitivity does not shrink with size (r = +0.20). Right: correlation of each index with yield anomalies; in the 240 severe country-years (SPEI-12 below −1.5) the six-month minimum tracks losses better than the annual index.

Diversity across the sample



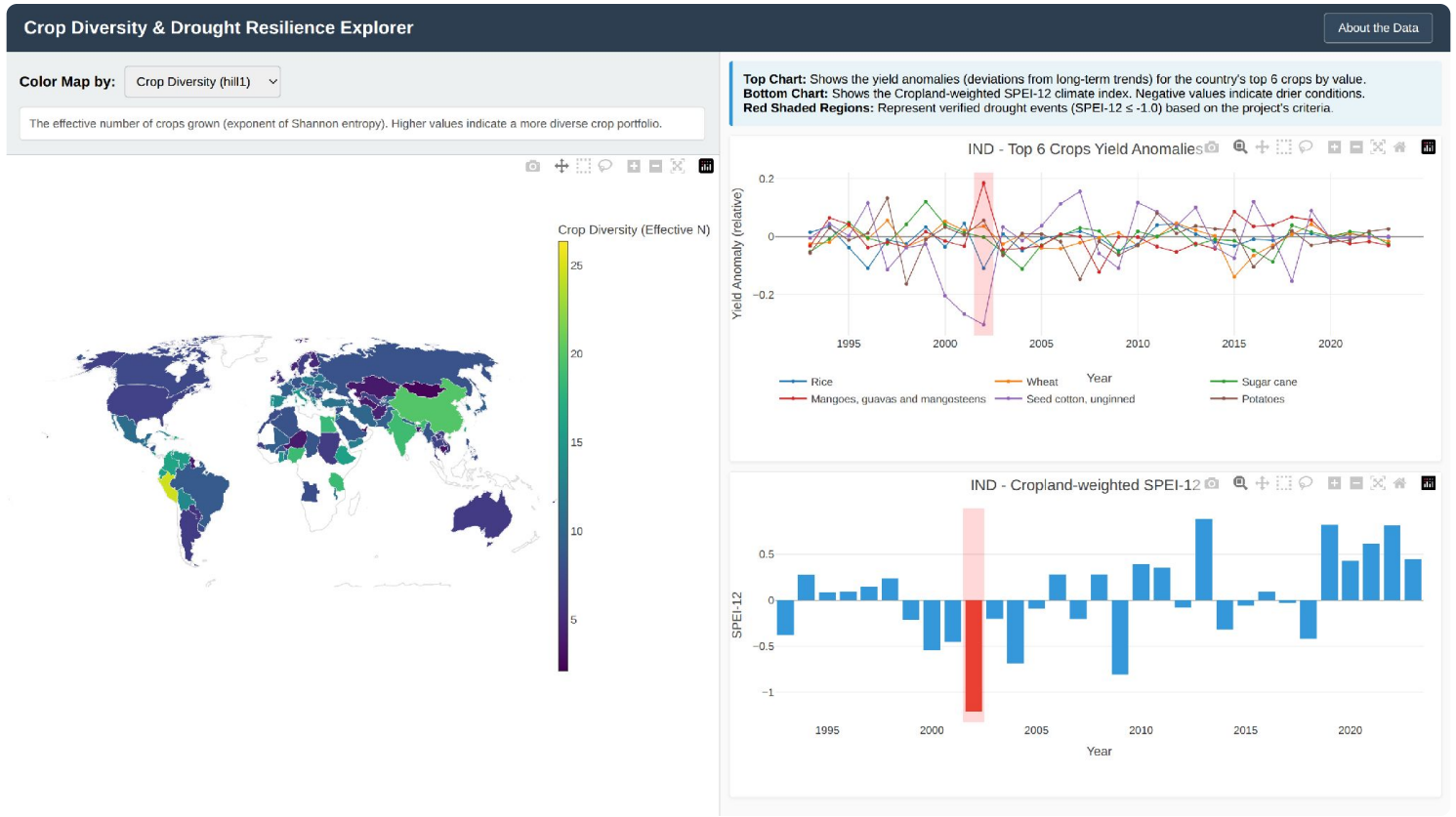
Source: FAOSTAT (QCL, QV), SPEIbase v2.10, MapSPAM 2020, World Bank WDI. 1993–2023, n = 107 countries.

9.2

median effective number of crops

Top map: crop diversity (hill1). Bottom map: response diversity (resp_div), plotted for 106 countries. Crop diversity ranges from about 2 to 26 effective crops: China, India and Peru are high, while Mongolia and much of Central Asia are low. Response diversity is highest in North Africa and Australia (scale capped at the 95th percentile). Grey countries did not pass the data-quality rules.

Interactive explorer



107

countries to open and compare

Pick a metric. The map recolours by crop diversity, response diversity, stability, resistance, recovery, vulnerability or synchrony.

Click a country. The right panels show the yield anomalies of its six largest crops and its cropland-weighted SPEI-12, with drought years shaded.

Screenshot: India. The cotton collapse lines up with the shaded 2002 drought year.

What the evidence shows

1

Diversity buys steadiness, not shock absorption

Stability rises (ρ +0.49) and bad years fall (−0.45) with crop diversity. Resistance (+0.07) and recovery (−0.17) in single droughts show no clear link.

2

The link survives controls

Holding income, fertiliser and irrigation constant, crop diversity still predicts stability (+0.37 SD) and vulnerability (−0.41 SD). Irrigation also lowers vulnerability in drought years (−0.32).

3

More varied crop responses did not help

Response diversity tracks the single most drought-sensitive crop ($r = 0.88$) and goes with more bad years (ρ +0.52). It marks an outlier crop, not balance.

4

How drought is measured matters

Averaging over large countries mutes the signal ($r = -0.48$), and in severe years the six-month index tracks losses better (0.23 vs 0.17). The cropland-weighted index flags 6 of 12 known droughts.

Future work

Seasonal drought events

Define events with the six-month index, which tracked severe losses better and may catch the known droughts we missed.

The 88 excluded countries

Test looser data rules or national statistics, so producers such as Indonesia and South Africa enter the sample.

A sturdier response metric

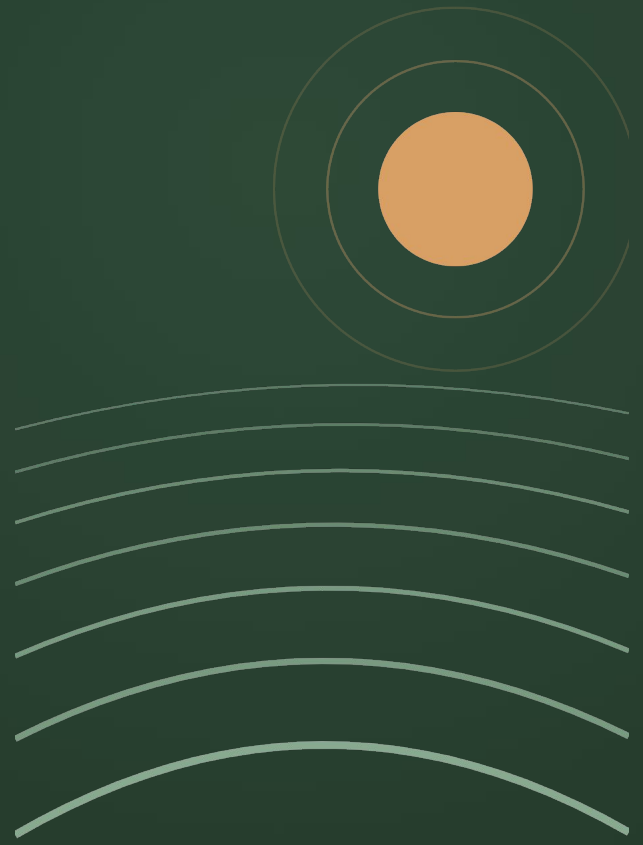
Separate real differences in crop response from one outlier crop and from noisy yield records.

Below the national scale

Repeat with state or district data, where averaging over large areas hides less of the drought.

Thank you

Questions are welcome :)



Data: FAOSTAT, SPEIbase v2.10, MapSPAM 2020, World Bank WDI, Natural Earth